Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 30 Days · Generated: 2026-05-06 07:30:01

Situation Summary

Stage IV: 5 of 26
Biomarker Count: 15 entries
Top Targets: ctDNA (5) · MSI/MMR (5) · KRAS (1)
Breakthrough: 7 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

HIGH Stage III ctDNA VERIFIED LIQUID BIOPSY PUBLISHED MAY 04, 2026 9

Circulating tumor DNA as a biomarker for the prediction of minimal residual disease and the individualization of adjuvant therapy in colorectal cancer

Circulating tumor DNA (ctDNA) shows potential as a biomarker for early detection of minimal residual disease (MRD) and for risk stratification in colorectal cancer (CRC). Its use may facilitate the individualization of adjuvant therapy. However, routine clinical implementation necessitates validation through ongoing randomized trials, particularly in the Czech context where ctDNA application is largely confined to research settings.

pubmed.ncbi.nlm.nih.gov

HIGH All Stages KRAS VERIFIED LIQUID BIOPSY PUBLISHED MAY 02, 2026 7

Translational feasibility of a plasmonic microarray-based liquid biopsy for KRAS codon mutation detection across tissue, plasma, and urine in early colorectal cancer

A 3D plasmonic microarray was developed for detecting KRAS codon mutations in colorectal cancer using blocked recombinase polymerase amplification and plasmon-enhanced fluorescence. This method improves sensitivity for rare ctDNA variants in early disease stages compared to conventional liquid biopsy techniques. The study highlights the potential of this approach for precision oncology in CRC.

pubmed.ncbi.nlm.nih.gov

HIGH Stage IV MSI/MMR VERIFIED LIQUID BIOPSY PUBLISHED MAY 01, 2026 7

Impact of treatment discontinuation among patients with microsatellite instability or mismatch repair-deficient metastatic colorectal cancer treated with immune checkpoint inhibitors

In patients with microsatellite instability or mismatch repair-deficient metastatic colorectal cancer treated with immune checkpoint inhibitors, early treatment discontinuation (<2 years) appears safe for those responding. Circulating tumor DNA (ctDNA) may help identify optimal stopping points. However, findings are based on a retrospective design with limited sampling, necessitating prospective studies to clarify ctDNA's role in guiding immune checkpoint inhibitor duration.

pubmed.ncbi.nlm.nih.gov

HIGH Stage IV ctDNA VERIFIED LIQUID BIOPSY PUBLISHED MAY 04, 2026 6

A New Era of Salvage-Line Treatment for Metastatic Colorectal Cancer: The Role and Clinical Significance of Circulating Tumor DNA

Longitudinal monitoring of circulating tumor DNA (ctDNA) is highlighted as a valuable tool for assessing dynamic molecular changes in metastatic colorectal cancer (mCRC). The study underscores the importance of comprehensive biomarker assessment in determining optimal treatment sequences amidst the expanding options of novel cytotoxic agents, multikinase inhibitors, and antibody-based therapies. This approach may enhance decision-making in salvage-line treatment for mCRC.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED RESEARCH PUBLISHED APR 13, 2026 6

Annual Review of Systemic Medical Treatment for Colorectal Cancer in 2025

Immunotherapy is now the standard first-line and perioperative treatment for mismatch repair-deficient or microsatellite instability-high colorectal cancer patients. Precision medicine has broadened to include BRAF V600E, KRAS mutations, and HER2 amplification. Circulating tumor DNA has been validated for surveillance and guiding rechallenge strategies.

pubmed.ncbi.nlm.nih.gov

HIGH Stage I BROAD SCOPE VERIFIED GUIDELINE PUBLISHED MAY 04, 2026 3

Truncating Adenomatous Polyposis Coli Mutations Serve as a Biomarker for Advanced Tumor Stage in Iranian Patients with Familial Colorectal Cancer

Truncating mutations in the Adenomatous Polyposis Coli (APC) gene's Mutation Cluster Region (MCR) were identified as biomarkers for advanced tumor stage in a cohort of 96 Iranian patients with familial colorectal cancer. The study utilized sequencing to classify variants according to ACMG guidelines, highlighting the prognostic significance of specific APC mutations in this population.

pubmed.ncbi.nlm.nih.gov

HIGH All Stages ctDNA VERIFIED LIQUID BIOPSY PUBLISHED APR 17, 2026 3

How Survivors of Colorectal Cancer Experience ctDNA Testing and Fear of Recurrence

In a cohort of colorectal cancer survivors, ctDNA-informed adjuvant chemotherapy (ACT) decisions did not increase fear of cancer recurrence (FCR) or cause long-term psychosocial harm, on average 5 years post-trial entry. The study highlights the importance of clear communication strategies as ctDNA testing becomes routine, ensuring patient understanding and trust in clinician-patient decision-making.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED RESEARCH PUBLISHED APR 14, 2026 3

Development and validation of a multicenter nomogram predicting the risk of liver metastasis after curative resection of colorectal cancer

A nomogram integrating clinicopathological features with KRAS, BRAF, and MSI status was developed and validated across multiple centers. This model demonstrated improved predictive accuracy and generalizability over conventional staging systems, aiding in the identification of high-risk colorectal cancer patients for tailored postoperative surveillance strategies to enhance long-term survival outcomes.

pubmed.ncbi.nlm.nih.gov

MEDIUM Stage I BRAF VERIFIED RESEARCH PUBLISHED APR 14, 2026 3

Increased Risk of Central Mesocolic Lymph Node Metastases in BRAF-Mutated Stage I-III Colon Cancer

BRAF mutation status was assessed in a prospective cohort of 97 stage I-III colon cancer patients from the T-REX trial. The study found that BRAF-mutated patients had an increased risk of central mesocolic lymph node metastases. This suggests that molecular profiling for BRAF mutations may guide decisions regarding the extent of lymphadenectomy, potentially improving surgical outcomes in this subset of patients.

pubmed.ncbi.nlm.nih.gov

MEDIUM Stage IV BROAD SCOPE VERIFIED RESEARCH PUBLISHED APR 22, 2026 2

Immune checkpoint inhibitors in *POLE/POLD1* proofreading-deficient CRC: from molecular basis to clinical practice and future directions

Pathogenic aberrations in the exonuclease domains of POLE/POLD1 define a hyper-immunogenic subset of metastatic colorectal cancer (mCRC) with an ultra-hypermutated phenotype. This distinct molecular profile suggests that immune checkpoint inhibitors (ICIs) may be effective in this population, expanding treatment options beyond the established dMMR/MSI-H cohort. Further exploration of this biomarker's role in clinical practice is warranted.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages MSI/MMR VERIFIED RESEARCH PUBLISHED APR 18, 2026 2

MSIanalyzer: Targeted Nanopore Sequencing Enables Single Nucleotide Resolution Analysis of Microsatellite Instability Diversity

MSIanalyzer utilizes targeted Oxford Nanopore sequencing to profile microsatellite instability (MSI) at single-nucleotide resolution, including base-level annotation of non-repeat interruptions. The study characterizes repeat-unit distributions in mismatch repair (MMR)-deficient and MMR-proficient cell lines and primary human tissues. This method enhances the understanding of MSI diversity, potentially informing therapeutic strategies in colorectal cancer.

pubmed.ncbi.nlm.nih.gov

MEDIUM Early Detection BROAD SCOPE VERIFIED EARLY DETECTION PUBLISHED APR 15, 2026 2

Immunohistochemical Expression of MLH1 and MSH2 in Colorectal Carcinoma and Its Correlation With Clinicopathological Parameters

MLH1 and MSH2 expression was assessed in colorectal carcinoma using immunohistochemistry to evaluate their correlation with clinicopathological parameters. The study highlights the significance of MMR deficiency in predicting outcomes, screening for Lynch syndrome, and guiding treatment decisions, particularly regarding immune checkpoint inhibitors.

pubmed.ncbi.nlm.nih.gov

Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the highest-scoring study per biomarker target drawn from the full research archive — not limited to the current report window. Some entries may predate the timeline above; they are included because they remain the most clinically significant reference available for that biomarker.

ctDNA
Circulating tumor DNA as a biomarker for the prediction of minimal residual disease and the individualization of adjuvant therapy in colorectal cancer - a review of current evidence and perspectives

Circulating tumor DNA (ctDNA) shows potential as a biomarker for early detection of minimal residual disease (MRD) and for risk stratification in colorectal cancer (CRC). Its use may facilitate the individualization of adjuvant therapy. However, routine clinical implementation necessitates validation through ongoing randomized trials, particularly in the Czech context where ctDNA application is largely confined to research settings.

TP53
Clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer

TP53 mutations serve as an independent prognostic marker for prolonged progression-free survival in metastatic colorectal cancer. While overall survival differences were not statistically significant, the findings suggest the need for further validation in prospective studies to assess TP53's role in therapeutic stratification.

BRAF
Detection of KRAS, NRAS and BRAF Mutations in Liquid Biopsy from Patients with Colorectal Cancer

ddPCR analysis of circulating tumor DNA (ctDNA) in colorectal cancer patients reveals that it can provide complementary information for molecular diagnosis. The study highlights the utility of liquid biopsy in detecting KRAS, NRAS, and BRAF mutations, enhancing diagnostic accuracy in this patient population.

KRAS
Translational feasibility of a plasmonic microarray-based liquid biopsy for KRAS codon mutation detection across tissue, plasma, and urine in early colorectal cancer

A 3D plasmonic microarray was developed for detecting KRAS codon mutations in colorectal cancer using blocked recombinase polymerase amplification and plasmon-enhanced fluorescence. This method improves sensitivity for rare ctDNA variants in early disease stages compared to conventional liquid biopsy techniques. The study highlights the potential of this approach for precision oncology in CRC.

MSI/MMR
Impact of treatment discontinuation among patients with microsatellite instability or mismatch repair-deficient metastatic colorectal cancer treated with immune checkpoint inhibitors

In patients with microsatellite instability or mismatch repair-deficient metastatic colorectal cancer treated with immune checkpoint inhibitors, early treatment discontinuation (<2 years) appears safe for those responding. Circulating tumor DNA (ctDNA) may help identify optimal stopping points. However, findings are based on a retrospective design with limited sampling, necessitating prospective studies to clarify ctDNA's role in guiding immune checkpoint inhibitor duration.

NGS
Exploratory biomarker findings from regorafenib plus toripalimab in patients with refractory metastatic colorectal cancer (REGOTORI study)

In the REGOTORI study, a multi-omics analysis identified several biomarkers associated with regorafenib plus toripalimab in refractory metastatic colorectal cancer. Key findings include ctDNA maxAF, HAF-bTMB, bITH, HAF-bITH, and mutational status of SMAD4 or PIK3CA, along with TME markers, which may serve as predictive or prognostic indicators for treatment response.

RAS
Effect of sex differences on the emergence of ctDNA RAS mutations in RAS wild-type colorectal cancer

In a single-center retrospective study of 43 patients with microsatellite stable RAS and BRAF wild-type metastatic colorectal cancer, the emergence of RAS mutations during chemotherapy was analyzed using a ctDNA assay. The study aimed to identify clinicopathological factors associated with this emergence, highlighting the potential impact of sex differences on RAS mutation development in RAS wild-type mCRC.

Genomic Profiling

Integrated Molecular Profiling of Colorectal Cancer by Tumor Location: Evidence from a Real-World Cohort with Primary and Metastatic Samples

Colorectal cancer (CRC) tumor location correlates with distinct molecular profiles. Right-sided tumors exhibit dMMR, MSI-H, and elevated TMB, while left-sided tumors show increased ERBB2 alterations and class 5 APC mutations. These findings underscore the necessity of incorporating tumor location into personalized molecular diagnostics and treatment strategies for CRC patients.

HER2

Targeting HER2 in Metastatic Colorectal Cancer: Current Therapies, Biomarker Refinement, and Emerging Strategies

HER2 overexpression/amplification occurs in 3-5% of metastatic colorectal cancers (mCRCs) and is a therapeutically actionable biomarker. The review discusses established and emerging HER2-targeted therapies, highlighting their incorporation into treatment guidelines based on increasing clinical evidence.

EGFR

Circulating Butyrate Attenuates Cetuximab Efficacy in Colorectal Cancer Through EGFR and AMPK-Wip1 Signaling

Butyrate may serve as a predictive biomarker for treatment response in KRAS wild-type colorectal cancer, as it reduces cetuximab efficacy via EGFR and AMPK-Wip1 signaling pathways.

miRNA

Diagnostic Performance of Circulating miRNA-92a and Related Circulating miRNA Panels for Colorectal Cancer: An Updated Systematic Review and Meta-Analysis

Circulating miRNA-92a shows superior diagnostic performance as a non-invasive biomarker for early colorectal cancer detection. The systematic review and meta-analysis suggest its potential utility, though further large-scale prospective studies are needed to standardize testing protocols and confirm clinical applicability.

IHC

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

Fused multiplex immunohistochemistry (mIHC)-derived radiomic and deep learning features provide accurate and interpretable predictions for multiple colorectal cancer outcomes. The study supports the integration of these features into precision oncology workflows, enhancing the potential for reproducible multi-outcome predictions across different centers.

Stool DNA

An algorithm-enhanced stool DNA system improves the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients

The FIT-sDNA-CA system, which integrates genetic, epigenetic, and inflammatory biomarkers, enhances the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients. This algorithm-enhanced stool DNA testing improves specificity compared to traditional fecal DNA tests, serving as an accurate non-invasive triage tool that aids in early risk stratification and reduces unnecessary endoscopic referrals.

What this means for you

If you have been diagnosed with colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* This is especially important if you have been diagnosed with metastatic (Stage IV) colorectal cancer, where biomarker results directly determine which targeted therapies and immunotherapies are available to you. Based on the research in this report, the most actively studied biomarkers right now are **ctDNA, TP53, BRAF, KRAS, and MSI/MMR**. Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision.

Generated from structured biomarker research data · 2026-05-06 07:30:01
This is not medical advice. Always speak with your oncologist before making any treatment decisions.

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

Colon & Colorectal Cancer — Biomarker Testing Intelligence

About This Report

This report is generated automatically from publicly available medical databases including PubMed, ASCO Journal of Clinical Oncology, Annals of Oncology, and NCI Cancer.gov. Entries are classified using deterministic rule-based logic and scored by clinical significance. AI-generated summaries are grounded exclusively in the source data provided.

Limitations

This report does not represent a complete picture of all published research. Database coverage, feed availability, and relevance filtering affect what appears. Research findings may not yet reflect current clinical guidelines. Always verify information at the original source before making clinical decisions.

Disclaimer

Alert classification fields are generated using deterministic rule-based analysis. Entry summaries are AI-generated from publicly available PubMed abstracts and society publications maintained by the U.S. National Library of Medicine and professional oncology societies. This report aggregates publicly available data for situational awareness only. No source content is reproduced verbatim. This report is made available under a free-use license for patient advocacy organizations, oncology practices, and individual patients. You may share or distribute it freely. You may not modify it, remove attribution, or republish it as your own work. Authorship and all intellectual property rights are retained by Randy Hinton. This report is provided for informational purposes only and does not constitute medical advice. Always consult your oncologist or a qualified healthcare provider before making any treatment decisions.

Copyright